

A top-down view of a wooden desk. In the top left is a small potted plant with green grass-like leaves. To its right is a white computer keyboard. In the bottom right is a white cup of dark coffee on a saucer. Below the keyboard is a black spiral-bound notebook with a pen resting on it. Two black binder clips are on the desk near the keyboard. The text 'Sampling-biased Performance Evaluation' is overlaid in white on the right side of the image.

Sampling-biased Performance Evaluation

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Contents

- **Sampling**
- **▯Types of Sampling**
- **▯Terminologies**
- **▯Introduction to Sampling Bias**
- **▯Types of Sampling Bias**
- **▯Performance Biased Sampling**
- **▯Real world example**
- **▯Mitigation Strategies**
- **▯Conclusion**
- **▯References**

Sampling

- Sampling in statistics refers to the process of selecting a subset of objects, individuals or items from a larger population in order to generate inference(s) or draw conclusion(s) about the entire population.
- This is done because often it is impractical, infeasible or extremely difficult to collect data from every single object or member of the population due to factors like time, cost, and accessibility.
- By studying a smaller sample, statisticians aim to make valid and accurate statements about the population, as a whole.

Types of Sampling

- **Random Sampling:** In this approach, every member of the population has an equal chance of being selected for the sample. This reduces bias and allows for generalization from the sample to the population. Simple random sampling and stratified random sampling are common approaches.
- **Stratified Sampling:** The population is divided into distinct subgroups or strata based on certain characteristics, and then random samples are taken from each stratum. This is useful when certain subgroups are of particular interest.
- **Systematic Sampling:** In this method, a fixed interval is chosen, and every n^{th} member of the population is selected for the sample. This can be efficient but might introduce bias if there's a pattern (which may not sync with the interval) in the data.

Types of Sampling

- **Cluster Sampling:** The population is divided into clusters or groups, and a random sample of clusters is selected. Then, data is collected from all members within the selected clusters. This is useful when it's difficult to access individuals directly.
- **Convenience Sampling:** This involves selecting individuals who are easiest to reach or readily available. While it is convenient, it can introduce bias since it may not be representative of the entire population.
- **Judgmental (or Purposive) Sampling:** In this method, the researcher selects individuals based on their judgment of who would be most informative for the study. This can be useful for qualitative research but can also introduce bias.
- **Snowball Sampling:** Commonly used in social research, this method starts with a small set of individuals and then additional participants are added/included based on referrals from the initial participants.

Terminologies

Bias:

- Any illogical or irrational preference/ privilege given (either intentionally or non-intentionally) to certain entities (or 'sample' here).
- It is also a tendency to ignore evidence that doesn't line up with that assumption

Performance:

- Ability to show the results generated for a snapshot of time after an act.

Introduction to Sampling Bias

- Sampling Bias is a phenomenon that occurs when the sample used for analysis does not accurately represent the entire population. This discrepancy can lead to distorted performance measurements and ultimately affect decision-making.
- Sampling bias arises due to the non-random selection of samples. It can lead to over-representation or under-representation of specific groups or characteristics, skewing the results.
- Sampling bias can distort performance metrics such as accuracy, precision, and recall. Biased metrics may mis-represent the true effectiveness of a model or system.

Types of Sampling Bias

- **Performance bias** : It occurs when the compared groups do not receive a similar amount of attention.
- **Detection bias**: Detection bias refers to systematic differences between groups in how outcomes are determined.
- **Attrition bias**: Attrition bias refers to systematic differences between groups in withdrawals from a study.
- **Reporting bias**: Reporting bias refers to systematic differences between reported and unreported findings, to get the desired result.
- **Other biases**: Survivorship bias, Self-selection bias, Exclusion and under-coverage bias, Pre-screening bias.

Performance Biased Sampling

- **Performance-biased sampling** is also known as **outcome-biased sampling** or **selective sampling** , is a potential source of bias in research and data analysis where the samples collected or included in the analysis are not representative of the entire population or dataset, but rather are influenced by the observed outcomes or results. This bias can lead to inaccurate or misleading conclusions.
- In performance-biased sampling, samples are selected or included in a way that disproportionately favors certain outcomes. This can happen unintentionally or intentionally, often because researchers or analysts are more likely to include samples that show interesting, extreme, or positive results. For example, in medical research, if only successful treatments are highlighted while unsuccessful cases are omitted, it can give a distorted view of the treatment's effectiveness.
- This bias can lead to overestimating the prevalence or impact of certain outcomes, making them seem more common or significant than they actually are. It can also result in a lack of understanding about the full range of outcomes and their potential causes.

Real-World Example 1: Tossing a Coin

- **Objective:** Conducting an experiment to test the fairness of a coin.
- **Expectation:** To determine if the coin is unbiased (50% chance of landing heads and 50% chance of landing tails).
- **Process:** Flip the coin 10 times and record the outcomes.
- **Outcomes:** 1.Heads 2.Tails 3.Heads 4.Heads 5.Tails 6.Heads 7.Heads 8.Heads 9.Heads 10.Heads
- **Conclusion:** The coin is biased towards landing heads, as it landed heads 9 out of 10 times.
- **Actual Conclusion:** The above conclusion might be misleading due to performance-biased sampling. In this example, we have only recorded the outcomes that support the idea of a biased coin (heads), while omitting the tails outcomes that contradict this notion. The actual bias of the coin might not be accurately represented in this small sample.
- **Mitigation:** To mitigate performance-biased sampling in this scenario, we would need to ensure that we include all outcomes (both heads and tails) in our analysis. By considering the entire dataset, we can get a more accurate understanding of whether the coin is truly biased or if the observed outcome was just due to chance.

Real-World Example 2: Clinical Trials and Publication Bias

- **Objective:** Clinical trial to test the effectiveness of a new drug for treating a specific medical condition.
- **Process:** The trial involves a large number of participants, some of whom are given the new drug while others are given a placebo.
- **Observation:** Participants receiving the new drug are showing positive improvements in their condition. However, participants receiving the placebo are not showing significant changes.
- **Actions:** The company decides to halt the trial prematurely and publish a report that highlights the positive outcomes seen in the group receiving the new drug.
- **Conclusion:** In this scenario, the company has engaged in performance-biased sampling. They selectively reported only the favorable outcomes of the trial while omitting the full dataset, which would have included the less favorable outcomes from the placebo group. As a result, the published report presents an overly positive view of the drug's effectiveness, potentially leading to biased conclusions about its benefits.
- *This type of bias can have serious implications. Researchers, healthcare professionals, and the public might base decisions on incomplete or misleading information, and the actual effectiveness and safety of the drug could be misunderstood.*

Mitigation Strategies

- **Use Random Sampling**
 - ☐ **Perform Oversampling**
 - ☐ **Stratified Sampling**
 - ☐ **Avoid Convenience Sampling**
 - ☐ **Include all outcomes**
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- **Robust Analysis**
 - **Transparent Reporting**
 - **Cross Validation**
 - **Pre-registration**
 - **Publication Ethics**

Conclusion

- Sampling bias has far-reaching implications for performance measurement and analysis. Awareness of its existence, coupled with appropriate strategies, is key to obtaining accurate insights from data.
- Performance-biased sampling can lead to distorted conclusions by emphasizing certain outcomes over others.
- By applying sound research practices, transparent reporting, and rigorous sampling techniques, researchers can mitigate this bias and produce more accurate and reliable findings that contribute to the advancement of knowledge.

References

[1] <https://www.alchemer.com/resources/blog/how-to-avoid-sampling-bias-in-research/>

[2] https://www.investopedia.com/terms/s/sample_selection_basis.asp

[3] <https://www.indeed.com/career-advice/career-development/biased-sample#:~:text=A%20biased%20sample%20occurs%20when,accurate%20representation%20of%20the%20population.>

[4] <https://www.questionpro.com/blog/sampling-bias/#:~:text=Sampling%20bias%20or%20a%20biased,bias%20is%20Presidential%20election%20voters>

[5] <https://catalogofbias.org/biases/performance-bias/>